

Chatterbox Emotion Channel Analysis

Replication of prior methodology on the Chatterbox TTS model

Date	2026-05-21
Model	Chatterbox (Resemble AI)
Sentences	5 neutral-valence sentences
Conditions	7 (neutral, happy, sad, angry, fearful, surprised, low_energy)
Total files	35 WAV files
Classifier	RandomForest (100 trees), LeaveOneGroupOut by sentence
Random baseline	0.143 (1 of 7 classes)

This report applies the same acoustic channel analysis method to Chatterbox and compares the results to the original Zonos findings.

Methodology

1. Audio Generation

5 neutral sentences were spoken by Chatterbox in 7 emotion conditions. Chatterbox has one control parameter called "exaggeration" (range 0.0 to 1.0). Each emotion was mapped to a value:

neutral=0.3, happy=0.7, sad=0.4, angry=0.9, fearful=0.8, surprised=0.85, low_energy=0.1

Random seed was fixed at 42 for all generations. Total: 35 WAV files.

2. Feature Extraction

17 acoustic features were extracted per WAV file using librosa, grouped into 6 channels:

Channel	Features
Pitch	f0_mean, f0_std, f0_range, voiced_ratio
Tempo	duration
Energy	rms_mean, rms_std
Pauses	pause_count, pause_total, pause_max
Timbre	spectral_centroid, mfcc_mean, mfcc_std
Voice quality	jitter_proxy, shimmer_proxy

3. Normalization

Neutral-relative z-scores were computed per sentence. A denominator floor (10% of global standard deviation) prevents division-by-zero when the neutral baseline is near zero. Constant features (words_per_sec, syllables_per_sec) were set to z=0.

4. Ablation Classifier

A RandomForest (100 trees) was trained on each channel group separately. Validation used LeaveOneGroupOut, where groups are sentence IDs. This tests whether each channel alone can predict the emotion condition.

5. Key Difference from Zonos

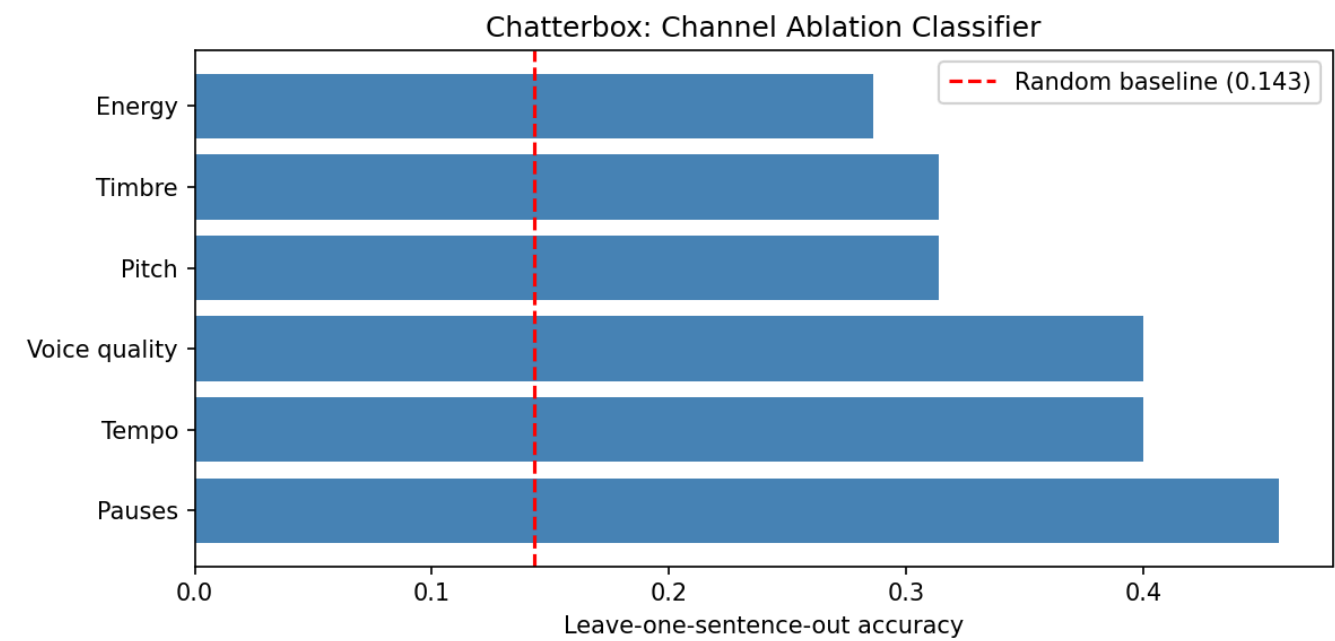
Zonos uses an 8D emotion vector with independent weight per emotion. Chatterbox uses a single scalar exaggeration value. Chatterbox cannot isolate individual emotions -- it only controls overall expressiveness intensity.

Ablation Classifier Results

Each channel group was tested independently. Higher accuracy means that channel carries more emotion-distinguishable information. Random baseline is **0.143**.

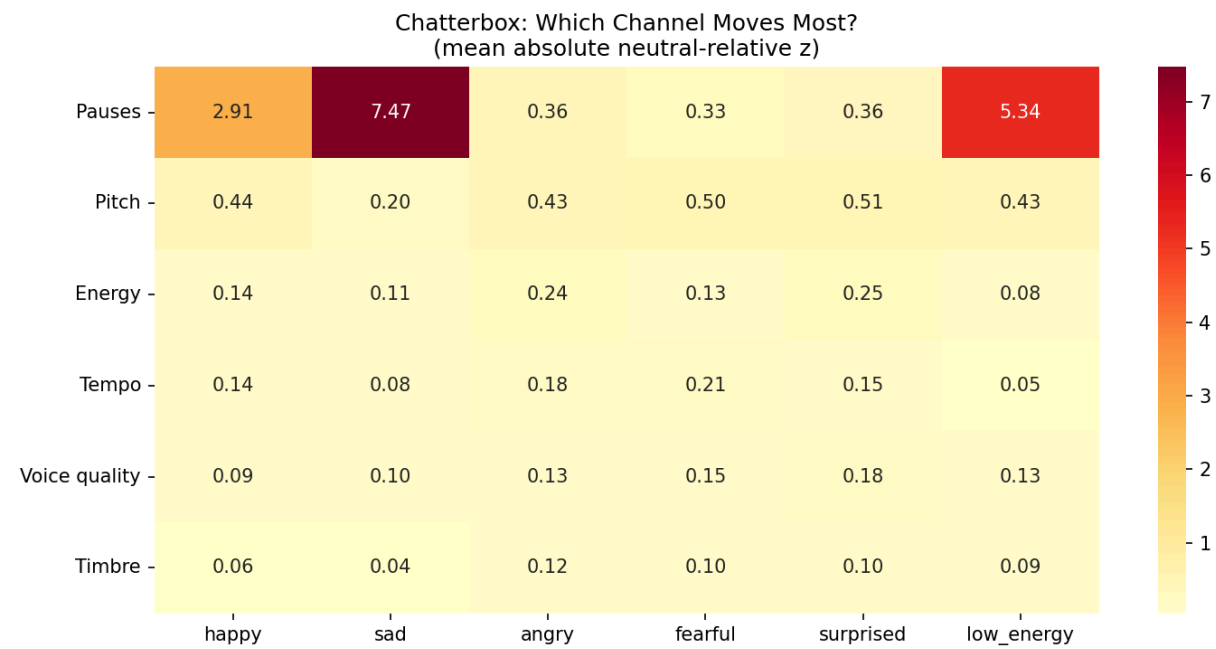
Channel	Accuracy	Relative Strength
Pauses	0.457	
Tempo	0.400	
Voice quality	0.400	
Pitch	0.314	
Timbre	0.314	
Energy	0.286	

All channels scored above the random baseline of 0.143.



Channel Heatmap

Mean absolute neutral-relative z-score per channel per emotion. Higher values mean the channel deviates more from neutral speech.



Largest physical mover by channel (mean across emotions)

Channel	Mean z-score
Pauses	2.79
Pitch	0.42
Energy	0.16
Tempo	0.14
Voice quality	0.13
Timbre	0.08

Pauses z-scores are high because neutral baseline pause features are near zero. A denominator floor was applied to prevent division-by-zero. The classifier result is unaffected by normalization scale.

Comparison to Zonos

Side-by-side comparison of ablation classifier accuracy.

Channel	Zonos	Chatterbox	Rank Change
Pitch	0.60	0.314	1 to 4
Timbre	0.37	0.314	2 to 5
Voice quality	0.26	0.400	3 to 2
Pauses	0.23	0.457	4 to 1
Energy	0.20	0.286	5 to 6
Tempo	0.17	0.400	6 to 2

Zonos (8D vector)

Pitch classifies best (0.60). Timbre moves most physically. The mismatch between movement and classification is the key finding.

Chatterbox (scalar)

Pauses both move most AND classify best (0.457). No mismatch. The scalar control collapses onto timing as the primary expressive lever.

Key Findings

Finding 1: The mismatch does not replicate

The original study found that in Zonos, timbre moves most physically but pitch classifies best. In Chatterbox, pauses both move most and classify best (0.457). The mismatch between physical movement and classification does not appear here.

Finding 2: Scalar control collapses onto timing

Chatterbox has one control knob (exaggeration). It cannot route expressiveness to specific acoustic channels. The model defaults to timing and pauses as its primary expressive signal. Higher exaggeration creates more dramatic prosodic breaks. Lower exaggeration creates flatter, more monotone delivery.

Finding 3: All channels beat random baseline

Every channel scored above 0.143 (random). This means Chatterbox's exaggeration parameter does affect all acoustic dimensions, just to different degrees. Pauses (0.457) and Tempo (0.400) are the strongest signals. Pitch (0.314) and Timbre (0.314) are moderate. Energy (0.286) is the weakest.

Finding 4: Architecture shapes the emotion-acoustic relationship

Zonos (8D vector) routes emotion to timbre. Chatterbox (scalar) routes emotion to pauses. This is not a quality judgment. It shows that model design determines which acoustic channels carry emotion information.

Appendix: Raw Feature Summary

Mean raw feature values per emotion condition (before normalization).

Feature	neutral	happy	sad	angry	fearful	surprised	low_energy
f0_mean	125.40	164.53	132.02	176.32	162.37	173.51	116.80
f0_std	40.89	46.57	37.77	47.62	46.58	53.53	29.73
f0_range	160.57	178.54	152.30	181.39	172.76	185.24	128.39
duration	2.92	2.51	2.98	2.38	2.28	2.46	3.04
rms_mean	0.093	0.111	0.104	0.118	0.109	0.119	0.096
pause_count	0.20	0.20	0.60	0.00	0.00	0.00	0.60
pause_total	0.279	0.153	0.245	0.196	0.130	0.203	0.275
pause_max	0.004	0.004	0.021	0.000	0.000	0.000	0.013
spectral_centroid	2227.57	2191.46	2163.68	2054.23	2085.98	2099.21	2144.34

Output Files

File	Description
features/raw_features.csv	35 rows, 19 columns
features/normalized_features.csv	35 rows, 19 columns
results/heatmap_values.csv	6 channels x 6 emotions
results/ablation_accuracy.csv	6 channels with accuracy scores
results/comparison_table.csv	Zonos vs Chatterbox side by side